

GROUP 1: Physical & Chemical Changes (Pretest)

Q1. Which of the following is a CHEMICAL change?

- A. Dissolving sugar in water
- B. Melting ice cream
- C. Cutting paper
- D. Rotting of tomatoes ✓

Answer: D – Rotting forms new substances; others only change appearance.

Q2. Which shows ONLY a PHYSICAL change?

- A. Crushing chunks of ice ✓
- B. Rusting of iron
- C. Burning wood
- D. Milk turning sour

Answer: A – Ice is still water; no new substance formed.

Q3. What is the MAIN difference between chemical and physical change?

- A. Physical change cannot be reversed
- B. Physical change always releases heat
- C. Chemical change produces a NEW substance ✓
- D. Chemical change is always faster

Answer: C – Composition changes in chemical change; stays the same in physical.

Q4. Which of these is a CHEMICAL change?

- A. Opening a soda can
- B. Photosynthesis in plants ✓
- C. Painting wood
- D. Bending metal

Answer: B – Plants turn CO₂ and water into glucose and oxygen.

Q5. Bubbles forming when vinegar is mixed with baking soda is evidence of:

- A. Evolution of gas ✓
- B. Color change
- C. Physical change only
- D. Precipitate formation

Answer: A – Bubbles indicate CO₂ gas is released.

Q6. Curdling of milk into lumpy solids is an example of which evidence?

- A. Temperature drop only
- B. Gas evolution
- C. Production of light
- D. Formation of a precipitate ✓

Answer: D – A precipitate is an insoluble solid formed during a reaction.

Q7. Mixing quicklime and water makes the container feel hot. What evidence is this?

- A. Color change
- B. Change in temperature ✓
- C. Gas formation
- D. Precipitate formation

Answer: B – Heat released indicates an exothermic chemical reaction.

Q8. Which is NOT evidence that a chemical reaction occurred?

- A. Production of light (like burning)
- B. Formation of gas bubbles
- C. Change in shape only by cutting ✓
- D. Change in color

Answer: C – Cutting changes only the shape, not the substance.

Q9. A student mixes two clear liquids and a yellow solid settles at the bottom. This is:

- A. Precipitate formation ☒
- B. Physical change
- C. Gas evolution
- D. Boiling

Answer: A – Formation of a solid from two liquids is a precipitate.

Q10. In a chemical equation, substances on the LEFT side of the arrow are called:

- A. Coefficients
- B. Products
- C. Precipitates
- D. Reactants ☒

Answer: D – Reactants are the starting substances.

Q11. Substances written on the RIGHT side of the arrow are:

- A. Symbols only
- B. Products ☒
- C. Reactants
- D. Gases always

Answer: B – Products are the new substances formed.

Q12. What does the symbol (aq) mean in a chemical equation?

- A. Solid precipitate
- B. Gaseous state
- C. Dissolved in water (aqueous) ☒
- D. Liquid pure substance

Answer: C – Aqueous means dissolved in water.

Q13. The symbol Δ written above the arrow means:

- A. Heat is supplied to the reaction ☒
- B. Gas is released
- C. A precipitate forms
- D. Reaction is reversible

Answer: A – Δ indicates heat is added.

Q14. The downward arrow ($\downarrow$) next to a product means:

- A. Heat released
- B. Gas evolved
- C. Liquid product
- D. A solid precipitate was formed ☒

Answer: D – The downward arrow indicates a precipitate.

Q15. What does the + sign between reactants mean?

- A. Heat is added
- B. Substances are mixed / added together ☒
- C. "Yields" or "produces"
- D. Only one reacts

Answer: B – The plus sign means "reacts with" or "and."

GROUP 1: Physical & Chemical Changes (Continued)

Q16. Which general equation represents COMBINATION (Synthesis)?

- A. $AB + CD \rightarrow AD + CB$
- B. $A + B \rightarrow AB$ ☒
- C. $AB \rightarrow A + B$
- D. $AB + C \rightarrow AC + B$

Answer: B – Two or more reactants form ONE product.

Q17. $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$ is an example of:

- A. Single displacement
- B. Combustion
- C. Decomposition
- D. Combination ☒

Answer: D – Na + Cl combine to make one product, NaCl.

Q18. The reaction $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ is classified as:

- A. Combination
- B. Double displacement
- C. Decomposition ☒
- D. Acid-base reaction

Answer: C – One reactant breaks down into two simpler substances.

Q19. What is the general form of DECOMPOSITION?

- A. $\text{AB} + \text{C} \rightarrow \text{AC} + \text{B}$
- B. $\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{Water}$
- C. $\text{A} + \text{B} \rightarrow \text{AB}$
- D. $\text{AB} \rightarrow \text{A} + \text{B}$ ☒

Answer: D – Opposite of combination; one becomes two or more substances.

Q20. $\text{Pb} + \text{FeSO}_4 \rightarrow \text{PbSO}_4 + \text{Fe}$ is which type?

- A. Single displacement ☒
- B. Double displacement
- C. Combustion
- D. Decomposition

Answer: A – Pb replaces Fe in the compound.

Q21. In SINGLE DISPLACEMENT, what happens?

- A. Ions of two compounds swap
- B. One element replaces another element in a compound ☒
- C. One compound splits into two
- D. Oxygen reacts with a hydrocarbon

Answer: B – General form: $\text{AB} + \text{C} \rightarrow \text{AC} + \text{B}$.

Q22. $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl}\downarrow + \text{NaNO}_3$ is:

- A. Decomposition
- B. Combination
- C. Double displacement ☒
- D. Single displacement

Answer: C – Ag^+ and Na^+ exchange partners.

Q23. Which is ALWAYS a reactant in COMBUSTION reactions?

- A. Water
- B. Nitrogen
- C. Carbon dioxide
- D. Oxygen gas (O_2) ☒

Answer: D – Combustion always requires oxygen.

Q24. Combustion of a hydrocarbon always produces these two products:

- A. Carbon dioxide (CO₂) and Water (H₂O) ☒
- B. Oxygen and hydrogen
- C. Salt and water
- D. Metal oxide and gas

Answer: A – Hydrocarbon + O₂ → CO₂ + H₂O.

Q25. HBr + NaOH → NaBr + H₂O is an example of:

- A. Combustion
- B. Acid-Base / Neutralization ☒
- C. Decomposition
- D. Single displacement

Answer: B – Acid + Base → Salt + Water.

Q26. Neutralization is a special type of:

- A. Decomposition
- B. Combination
- C. Combustion
- D. Double displacement ☒

Answer: D – H⁺ from the acid and the metal from the base exchange places.

Q27. Which change is reversible and only alters form?

- A. Digesting food
- B. Tarnishing silver
- C. Melting wax ☒
- D. Burning wood

Answer: C – Melting is a physical change.

Q28. Which observation does NOT suggest a chemical change?

- A. Permanent color change
- B. Light and heat given off
- C. Salt dissolving in water ☒
- D. New solid not easily separated

Answer: C – Dissolving salt is a physical change.

Q29. In the equation Mg + O₂ → MgO, what is magnesium oxide?

- A. Catalyst
- B. Product ☒
- C. Gas
- D. Reactant

Answer: B – It is on the right side of the arrow.

Q30. Which reaction type is: Fuel + O₂ → CO₂ + H₂O?

- A. Decomposition
- B. Synthesis
- C. Single displacement
- D. Combustion ☒

Answer: D – Burning a fuel in oxygen is combustion.

Q31. What does the arrow (→) mean in a chemical equation?

- A. Reacts with

- B. Yields / produces ☒
- C. Dissolves in
- D. Equals

Answer: B – The arrow separates reactants from products.

Q32. Which is a decomposition reaction?

- A. $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$
- B. $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- C. $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ ☒
- D. $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

Answer: C – One reactant breaks into two products.

Q33. A reaction where two compounds exchange ions is called:

- A. Double displacement ☒
- B. Single displacement
- C. Decomposition
- D. Synthesis

Answer: A – The ions swap partners.

Q34. Which is an example of synthesis?

- A. Sodium chloride + Silver nitrate $\rightarrow$ Silver chloride + Sodium nitrate
- B. Water $\rightarrow$ Hydrogen + Oxygen
- C. Iron + Sulfur $\rightarrow$ Iron sulfide ☒
- D. Wood + Oxygen $\rightarrow$ Ash + CO_2

Answer: C – Two elements combine to form one compound.

Q35. Which symbol shows a gaseous product?

- A. (l)
- B. (aq)
- C. (s)
- D. (g) or $\uparrow$ ☒

Answer: D – (g) or an upward arrow indicates a gas.

GROUP 3 · ACIDS, BASES, AND SALTS (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

GROUP 2: Acids, Bases, and Salts (Randomized Answers)

Q1. Which is a property of ALL acids?

- A. Bitter taste
- B. Soapy feel
- C. $\text{pH} > 7$
- D. Sour taste; turns blue litmus RED ☒

Answer: D – Classic acid property (vinegar, calamansi).

Q2. Bases turn red litmus paper:

- A. Yellow
- B. Blue ☒
- C. Colorless
- D. Red

Answer: B – Mnemonic: Base turns Red litmus Blue.

Q3. Pure water at 25°C has pH:

- A. 14
- B. 5
- C. 7 ☒
- D. 0

Answer: C – pH 7 is neutral.

Q4. Phenolphthalein is colorless in acid and _____ in base.

- A. Green
- B. Yellow
- C. Bright pink ☒
- D. Red

Answer: C – Standard indicator rule.

Q5. Which ion do acids release in water?

- A. Cl^-
- B. Na^+
- C. OH^-
- D. H^+ / H_3O^+ ☒

Answer: D – Arrhenius definition of an acid.

Q6. Bases release which ion in water?

- A. O^{2-}
- B. OH^- (hydroxide) ☒
- C. H_3O^+
- D. H^+

Answer: B – Arrhenius definition of a base.

Q7. Common table salt (saline solution) is chemically:

- A. NaOH
- B. KCl
- C. NaCl ☒
- D. HCl

Answer: C – Sodium chloride is a neutral salt.

Q8. Hydrochloric acid found in the stomach has formula:

- A. CH_3COOH
- B. HCl ☒
- C. H_2SO_4
- D. HNO_3

Answer: B – HCl is gastric acid.

Q9. Sodium hydroxide, a common strong base (lye), has formula:

- A. NH_4OH
- B. NaCl
- C. NaOH ☒
- D. $\text{Ca}(\text{OH})_2$

Answer: C – NaOH is caustic soda.

Q10. Methyl orange is _____ in acid and _____ in base.

- A. Blue / Red

- B. Red / Yellow ☒
- C. Yellow / Red
- D. Pink / Colorless

Answer: B – Standard synthetic indicator.

Q11. Which is a natural indicator?

- A. Litmus paper only
- B. Turmeric ☒
- C. Phenolphthalein
- D. Methyl orange

Answer: B – Turmeric, red cabbage, and onion are natural indicators.

Q12. A solution with pH 7 is:

- A. Basic
- B. Strongly acidic
- C. Neutral ☒
- D. Acidic

Answer: C – Equal H^+ and OH^- concentration.

Q13. Which acid is found in vinegar?

- A. Sulfuric acid
- B. Acetic acid ☒
- C. Citric acid
- D. HCl

Answer: B – Acetic acid gives vinegar its sour taste.

Q14. Which base is found in antacids like milk of magnesia?

- A. NaOH
- B. NH_4OH
- C. KOH
- D. $Mg(OH)_2$ ☒

Answer: D – Neutralizes excess stomach acid.

Q15. When blue litmus paper is placed in a base, it stays:

- A. Green
- B. Blue ☒
- C. Colorless
- D. Red

Answer: B – Blue litmus remains blue in a base.

Q16. Which set are ALL ACIDS?

- A. Soap, detergent, antacid
- B. NaOH, KOH, $Ca(OH)_2$
- C. HCl, vinegar, calamansi juice ☒
- D. NaCl, baking soda, bleach

Answer: C – All are acids.

Q17. Which set are ALL BASES?

- A. Salt, sugar, water
- B. NaOH, bleach, milk of magnesia ☒
- C. Vinegar, lemon, soda

D. HCl, HNO₃, H₂SO₄

Answer: B – All are bases.

Q18. Antacids (Kremil-S) fight hyperacidity because they are:

- A. Neutral salts
- B. Strong acids
- C. Indicators
- D. Weak bases ☒

Answer: D – Mild bases neutralize stomach acid.

Q19. A solution with pH = 4 is:

- A. Neutral
- B. Strongly acidic ☒
- C. Weak base
- D. Weak acid

Answer: B – Lower pH means stronger acidity.

Q20. pH = 11 is classified as:

- A. Neutral
- B. Acidic
- C. Strongly basic / alkaline ☒
- D. Weakly acidic

Answer: C – High pH indicates a strong base.

Q21. Which pH is the WEAKEST acid?

- A. pH 8
- B. pH 3
- C. pH 6 ☒
- D. pH 1

Answer: C – Closest to 7 on the acidic side.

Q22. Acid + Base → Salt + Water. This reaction is:

- A. Combination
- B. Decomposition
- C. Neutralization ☒
- D. Combustion

Answer: C – Neutralization cancels acidic and basic properties.

Q23. Turmeric stain turns reddish-brown when soap is applied because soap is:

- A. Basic ☒
- B. Neutral
- C. Acidic
- D. A salt

Answer: A – Turmeric turns reddish-brown in a base.

Q24. Onion and vanilla change SMELL in acids/bases. They are:

- A. Catalysts
- B. Olfactory indicators ☒
- C. Synthetic indicators
- D. Natural color indicators

Answer: B – They indicate acidity/basicity through smell.

Q25. Tooth decay starts when mouth pH:

- A. Equals 7
- B. Is exactly 9
- C. Drops below 5.5 ☒
- D. Rises above 8

Answer: C – Enamel begins dissolving below pH 5.5.

Q26. Which salt is basic when dissolved in water?

- A. NH_4Cl
- B. KNO_3
- C. NaCl
- D. Na_2CO_3 ☒

Answer: D – Sodium carbonate forms a basic solution.

Q27. Which salt is acidic when dissolved in water?

- A. KCl
- B. NH_4Cl ☒
- C. CH_3COONa
- D. Na_2SO_4

Answer: B – Ammonium chloride forms an acidic solution.

Q28. Dry HCl gas does NOT change dry blue litmus. Why?

- A. Litmus only works in bases
- B. Dry HCl is a base
- C. HCl is not an acid
- D. No water $\rightarrow$ no free H^+ ions ☒

Answer: D – Water is needed for HCl to ionize.

Q29. Each pH unit = _____ change in H^+ concentration.

- A. 100×
- B. 10× ☒
- C. 5×
- D. 2×

Answer: B – The pH scale is logarithmic.

Q30. Arrange STRONGEST ACID $\rightarrow$ STRONGEST BASE: pH 2, pH 12, pH 6, pH 7, pH 9.

- A. $12 \rightarrow 9 \rightarrow 7 \rightarrow 6 \rightarrow 2$
- B. $2 \rightarrow 7 \rightarrow 6 \rightarrow 9 \rightarrow 12$
- C. $2 \rightarrow 6 \rightarrow 7 \rightarrow 9 \rightarrow 12$ ☒
- D. $7 \rightarrow 2 \rightarrow 6 \rightarrow 9 \rightarrow 12$

Answer: C – Lowest pH is the strongest acid.

Q31. Which salt comes from a strong acid + strong base (neutral pH $\approx$ 7)?

- A. Sodium carbonate
- B. Sodium chloride (NaCl) ☒
- C. Ammonium chloride
- D. Sodium acetate

Answer: B – $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$.

Q32. Acidic/basic solutions conduct electricity because they have:

- A. Neutral molecules
- B. Metal atoms
- C. Mobile ions ☒
- D. Free electrons

Answer: C – Ions carry electric current.

Q33. When diluting concentrated acid in the lab, ALWAYS:

- A. Heat first
- B. Pour acid SLOWLY into water with stirring ☒
- C. Pour water into acid
- D. Mix fast in any order

Answer: B – This prevents dangerous splashing.

Q34. Which is a WEAK acid found in vinegar?

- A. H_2SO_4
- B. HCl
- C. CH_3COOH (acetic acid) ☒
- D. HNO_3

Answer: C – Acetic acid is the acid in vinegar.

Q35. Baking soda (NaHCO_3) reacts with acids to release:

- A. O_2
- B. N_2
- C. CO_2 gas ☒
- D. H_2

Answer: C – Acid + bicarbonate $\rightarrow$ salt + H_2O + CO_2 .

Here's Group 2 with the correct answers randomized (no obvious pattern). I only changed the order of the choices while keeping the questions and explanations the same.

Group 2 · TYPES OF CHEMICAL REACTIONS IN THE ENVIRONMENT

1. $\text{A} + \text{B} \rightarrow \text{AB}$ is:

- A. Single displacement
- B. Combustion
- C. Combination / Synthesis ☒
- D. Decomposition

2. $\text{AB} \rightarrow \text{A} + \text{B}$ is:

- A. Combination
- B. Double displacement
- C. Neutralization
- D. Decomposition ☒

3. $\text{Hydrocarbon} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{heat}$ is:

- A. Combustion ☒
- B. Acid base
- C. Precipitation
- D. Decomposition

4. $\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{Water}$ is:

- A. Decomposition
- B. Single replacement
- C. Neutralization (special double displacement) ☒
- D. Combustion

5. $\text{AB} + \text{C} \rightarrow \text{AC} + \text{B}$ is:

- A. Synthesis
- B. Decomposition
- C. Single Displacement / Replacement ☒
- D. Double displacement

6. $\text{AB} + \text{CD} \rightarrow \text{AD} + \text{CB}$ is:

- A. Combination

B. Double Displacement / Replacement ☒

C. Combustion

D. Single displacement

7. $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$ is:

A. Combustion

B. Single displacement

C. Combination ☒

D. Decomposition

8. $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ is:

A. Acid base

B. Combination

C. Double displacement

D. Decomposition ☒

9. Plants build glucose from $\text{CO}_2 + \text{H}_2\text{O}$ using sunlight. This is:

A. Corrosion

B. Photosynthesis (combination / endothermic) ☒

C. Respiration

D. Combustion

10. $\text{Glucose} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{energy}$ in body cells is:

A. Cellular Respiration (exothermic) ☒

B. Photosynthesis

C. Decomposition

D. Combustion only

11. Which reaction involves one reactant breaking down into many?

A. Neutralization

B. Single replacement

C. Decomposition ☒

D. Synthesis

12. Rusting of iron is an example of:

A. Photosynthesis

B. Neutralization

C. Combustion

D. Oxidation / Corrosion ☒

13. Precipitation reactions form:

A. Insoluble solid ☒

B. Gas only

C. Water only

D. No new substance

14. In a single displacement reaction:

A. All substances combine

B. One element swaps place with another ☒

C. One compound splits

D. Nothing changes

15. Photosynthesis is the reverse of:

A. Acid rain

B. Combustion

C. Respiration ☒

D. Decomposition

16. $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\uparrow$ is:

A. Combination

B. Decomposition

C. Double displacement

D. Single displacement (metal replaces H in acid) ☒

17. $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl}\downarrow + \text{NaNO}_3$ is:

A. Combustion

B. Synthesis

C. Double displacement + precipitation ☒

D. Single displacement

18. $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{heat}$ is:

A. Neutralization

B. Decomposition

C. Photosynthesis

D. Combustion of methane (natural gas) ☒

19. $\text{SO}_2 + \text{NO}_x$ from vehicles + rain $\rightarrow$

A. Ozone

B. Freshwater

C. Acid rain ☒

D. Neutral water

20. $4\text{Fe} + 3\text{O}_2 + \text{H}_2\text{O} \rightarrow 2\text{Fe}_2\text{O}_3 \cdot \text{H}_2\text{O}$ (rust) is called:

A. Corrosion / slow oxidation ☒

B. Combustion

C. Neutralization

D. Single displacement

21. Which reaction is OPPOSITE of photosynthesis?

A. Neutralization

B. Corrosion

C. Acid rain

D. Cellular respiration ☒

22. Limestone statues dissolve in acid rain via:

A. Photosynthesis

B. Acid Carbonate double displacement ☒

C. Combustion

D. Synthesis

23. Burning gasoline, forests, volcanoes all add what to atmosphere?

A. Water only

B. Nitrogen only

C. CO_2 (greenhouse gas) + acid rain gases ☒

D. Oxygen

24. How does photosynthesis help fight global warming?

A. Makes acid rain

B. Produces heat

C. Releases CO_2

D. Absorbs / removes CO_2 ; releases O_2 ☒

25. A reaction has only ONE reactant. It MUST be:

A. Single displacement

B. Decomposition ☒

C. Combination

D. Combustion

26. Burning magnesium ribbon: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ is:

A. Neutralization

B. Combination + oxidation ☒

C. Double displacement

D. Decomposition

27. Which reaction type always produces heat?

A. Electrolysis

B. Melting

C. Combustion ☒

D. Photosynthesis

28. $2\text{H}_2\text{O} \rightarrow 2\text{H}_2\uparrow + \text{O}_2\uparrow$ (with electricity) is:

A. Combustion

B. Single displacement

C. Combination

D. Electrolytic Decomposition ☒

29. $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$. Why does Fe replace Cu?

A. Cu more active

B. Random

C. Fe is MORE reactive (activity series rule) ☒

D. Fe is lighter

30. Which environmental reaction is BOTH combination AND exothermic?

A. Corrosion only

B. Electrolysis

C. Formation of acid rain gases in atmosphere ☒

D. Photosynthesis

31. Acid rain reacting with steel bridges is which type?

A. Decomposition

B. Single displacement (acid + metal) ☒

C. Combustion

D. Precipitation

32. A reaction has O_2 as reactant AND releases heat. It is almost always:

A. Combustion ☒

- B. Neutralization
- C. Decomposition
- D. Precipitation

33. Which correctly classifies respiration?

- A. Single replacement
- B. Endothermic decomposition
- C. Exothermic combustion-like oxidation ☒
- D. Endothermic combination

34. The main environmental impact of uncontrolled combustion is:

- A. Only corrosion
- B. Global warming + acid rain + air pollution ☒
- C. Ozone layer only
- D. Only photosynthesis increase

35. CaCO_3 (coral/shells) + ocean acidification causes:

- A. Respiration decrease
- B. Coral growth
- C. Photosynthesis
- D. Double displacement dissolving coral skeletons ☒

This version avoids the repetitive "B" pattern by distributing the correct answers across A, B, C, and D while keeping the content unchanged.

Group 4 · CHEMICAL EQUATIONS (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

1. Left side of the arrow ($\rightarrow$) contains:

- A. Products B. Reactants C. Coefficients D. Catalysts

☒ B — Starting materials you put in.

2. Right side of the arrow contains:

- A. Reactants B. Products C. Subscripts D. Heat

☒ B — New substances formed.

3. (aq) in equations means:

- A. Gas B. Pure liquid C. Aqueous / dissolved in water D. Solid

☒ C — From Latin aqua = water.

4. (s) means:

- A. Gas B. Solid C. Solution D. Catalyst

☒ B — Solid state of matter.

5. (g) means:

- A. Gas B. Liquid C. Aqueous D. Precipitate

☒ A — Gaseous state.

6. Law of Conservation of Mass says:

- A. Mass can be created B. Mass can be destroyed C. Total mass reactants = total mass products; atoms rearranged only D. Mass always increases

☒ C — Lavoisier's fundamental law.

7. The arrow $\rightarrow$ is read as:

- A. equals B. yields / produces / forms C. plus D. reacts with

☒ B — Standard chemical equation terminology.

8.

o between reactants is read as:

- A. yields B. reacts with / and C. produces D. equals

☒ B — Separates substances being mixed.

9. Balanced equations have _____ atoms of each element on both sides.

- A. different B. equal C. increasing D. random

☒ B — Required by conservation of mass.

10. Word equation: Hydrogen + Oxygen $\rightarrow$ Water. Formula version:

- A. $\text{H} + \text{O} \rightarrow \text{HO}$ B. $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ C. $\text{H}_2 + \text{O} \rightarrow \text{H}_2\text{O}$ D. $\text{H}_2\text{O} \rightarrow \text{H}_2 + \text{O}_2$

☒ B — Correct formulas + balanced.

11. Symbol (l) stands for:

- A. Solid B. Liquid C. Gas D. Dissolved

☒ B — Pure liquid state.

12. A coefficient tells you:

- A. Number of atoms in a molecule B. Number of molecules / moles of substance C. Charge of the ion D. Type of reaction

☒ B — Multiplies every atom in that formula.

13. A subscript tells you:

- A. Number of molecules B. Number of that specific atom in one molecule C. Mass of the substance D. Speed of reaction

✓ B — $\text{H}_2\text{O} = 2 \text{ H}, 1 \text{ O}$ per molecule.

14. Which is the correct formula for carbon dioxide?

A. CO B. CO_2 C. C_2O D. CO_3

✓ B — One carbon, two oxygen atoms.

15. In CaCl_2 , the subscript “2” refers to:

A. Calcium atoms B. Chlorine atoms C. Total molecules D. Charge

✓ B — Two chlorine atoms bonded to one calcium.

16. When balancing, you may ONLY change:

A. Subscripts B. Coefficients (big front numbers) C. Formulas D. Element symbols

✓ B — Changing subscripts changes the substance itself!

17. In $3 \text{ H}_2\text{O}$, “3” is a _____; small “2” is a _____.

A. subscript / coefficient B. coefficient / subscript C. reactant / product D. symbol / formula

✓ B — Coefficient $\times$ subscript = total atoms.

18. Which is already BALANCED?

A. $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$ B. $\text{Ca} + \text{Cl}_2 \rightarrow \text{CaCl}_2$ C. $\text{Na} + \text{Cl}_2 \rightarrow \text{NaCl}$ D. $\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$

✓ B — 1 Ca, 2 Cl each side.

19. In $2 \text{ H}_2\text{SO}_4$, total H atoms = ?

A. 2 B. 4 C. 6 D. 8

✓ B — $2 \times 2 = 4$ H atoms total.

20. In $4 \text{ Al}_2\text{O}_3$, total O atoms = ?

A. 3 B. 7 C. 12 D. 8

✓ C — $4 \times 3 = 12$ O atoms.

21. In $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$, how many Cl on product side?

A. 1 B. 2 C. 3 D. 4

✓ B — Subscript 2 on Cl in MgCl_2 .

22. First step in balancing any equation:

A. Add coefficients randomly B. Change subscripts C. Count atoms of each element on BOTH sides D. Add water

✓ C — Know what you have before adjusting.

23. Why must equations be balanced?

A. Teacher rule B. To obey Law of Conservation of Mass C. To look nice D. To speed reaction

✓ B — Atoms are never created/destroyed.

24. Open beaker releases gas; measured mass drops. Conservation law?

A. Broken B. Still holds — gas escaped; total mass of all matter unchanged C. Only applies in labs D. Atoms destroyed

✓ B — Closed system = same mass; open = gas escapes measurement.

25. $\text{N}_2 + \text{H}_2 \rightarrow \text{NH}_3$. Which atoms are unbalanced?

A. Only N B. Only H C. Both N and H D. Both balanced

✓ C — $2\text{N} \rightarrow 1\text{N}$; $2\text{H} \rightarrow 3\text{H}$.

26. In $5\text{C}_6\text{H}_{12}\text{O}_6$, total C atoms = ?

A. 6 B. 12 C. 30 D. 5

✓ C — $5 \times 6 = 30$ carbon atoms.

27. Which action is FORBIDDEN when balancing?

A. Adding coefficients B. Changing subscripts C. Counting atoms D. Writing formulas

✓ B — Alters the identity of the compound.

28. Balance: ___ H_2 + ___ $\text{O}_2 \rightarrow$ ___ H_2O

A. 1,1,2 B. 2, 1, 2 C. 2,2,2 D. 1,2,1

✓ B — $4\text{H} + 2\text{O}$ each side.

29. Balance: ___ N_2 + ___ $\text{H}_2 \rightarrow$ ___ NH_3

A. 1,2,2 B. 1,3,3 C. 1, 3, 2 D. 2,3,2

✓ C — $2\text{N}, 6\text{H}$ each side.

30. Balance: ___ Al + ___ $\text{O}_2 \rightarrow$ ___ Al_2O_3

A. 2,3,1 B. 2,1,2 C. 4, 3, 2 D. 3,4,2

✓ C — $4\text{Al}, 6\text{O}$ each side.

31. Balance: ___ Fe + ___ $\text{H}_2\text{O} \rightarrow$ ___ Fe_3O_4 + ___ H_2

A. 1,1,1,1 B. 3, 4, 1, 4 C. 2,3,1,2 D. 3,2,1,2

✓ B — $3\text{Fe}, 8\text{H}, 4\text{O}$ each side.

32. In balanced $2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$, total O atoms LEFT?

A. 7 B. 14 C. 8 D. 6

✓ B — $7 \times 2 = 14$ O in 7O_2 .

33. If 12g C reacts completely with 32g O_2 , mass of CO_2 formed = ?

A. 12g B. 32g C. 44g D. 20g

✓ C — $12 + 32 = 44\text{g}$; conservation of mass.

34. What is the BEST explanation for balanced equations?

A. Atoms multiply B. Atoms are rearranged; same number before and after C. New atoms form D. Atoms become smaller

✓ B — Core meaning of conservation of mass.

35. Never change subscripts because:

A. It makes numbers bigger B. It changes the identity of the substance C. It violates charge D. Coefficients are easier

✓ B — H_2O vs H_2O_2 = completely different chemicals.

Group 5 · CHEMICAL EQUATIONS EQUATIONS (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

1. In a chemical equation, what do we call the substances that are present BEFORE the reaction takes place?

A. Catalysts

B. Products

C. Reactants

D. Precipitates

✓ C — Reactants are the starting materials; they appear on the left side of the arrow.

2. Where are the PRODUCTS of a chemical reaction written in an equation?

A. Above the arrow

B. Left side of the arrow

C. Right side of the arrow

D. Below the arrow

✓ C — Products are the new substances formed; they appear after the reaction arrow.

3. What does the symbol " $\rightarrow$ " mean when used in a chemical equation?

A. Dissolves in

B. Reacts to form

C. Mixes with

D. Equals

✓ B — It shows the direction of the chemical change from reactants to products.

4. Which symbol indicates that a substance is a SOLID in a chemical equation?

A. (g)

B. (aq)

C. (l)

D. (s)

✓ D — State symbols tell the physical form of each substance; (s) = solid.

5. The symbol (aq) next to a substance means:

A. It is dissolved in water

B. It is a gas

C. It is a solid powder

D. It is pure liquid water

✓ A — "Aqueous" means dissolved in water to form a solution.

6. A number written IN FRONT of a chemical formula (e.g., " $2\text{H}_2\text{O}$ ") is called a:

A. Charge

B. Superscript

C. Coefficient

D. Subscript

✓ C — Coefficients show how many molecules or formula units take part in the reaction.

7. A small number written AFTER an element symbol (e.g., " O_2 ") is called a:

A. Mass number

B. Coefficient

C. Yield

D. Subscript

✓ D — Subscripts show how many atoms of that element are in one molecule or formula unit.

8. What does the law of conservation of mass state about chemical reactions?

A. Total mass increases when new substances form

B. Mass changes depending on temperature

C. Total mass of reactants equals total mass of products

D. Total mass decreases when bonds break

✓ C — Atoms are only rearranged; no atoms are created or destroyed.

9. A reaction where two or more substances combine to make ONE new substance is called:

A. Double displacement

B. Single displacement

C. Combination / Synthesis

D. Decomposition

✓ C — General form: $\text{A} + \text{B} \rightarrow \text{AB}$.

10. A reaction where ONE reactant breaks apart into two or more simpler substances is:

A. Combustion

B. Decomposition

C. Replacement

D. Synthesis

✓ B — General form: $AB \rightarrow A + B$.

11. What is a CATALYST?

A. A gas always produced

B. A product formed at the end

C. A substance used up during the reaction

D. A substance that speeds up a reaction without being changed or used up

✓ D — Catalysts are often written above the reaction arrow.

12. A reaction that releases heat or light energy to the surroundings is called:

A. Exothermic

B. Thermal

C. Endothermic

D. Isothermic

✓ A — Energy comes OUT of the reaction.

13. A reaction that ABSORBS energy from its surroundings (feels cold) is:

A. Combustion

B. Endothermic

C. Displacement

D. Exothermic

✓ B — Energy goes INTO the reaction.

14. In a single displacement reaction:

A. One element replaces another element in a compound

B. Nothing changes partners

C. One compound breaks into elements

D. Two compounds swap partners

✓ A — General form: $A + BC \rightarrow AC + B$.

15. In a double displacement reaction:

A. A fuel reacts with oxygen

B. One element replaces another

C. Two compounds exchange ions to form new compounds

D. Two elements combine

✓ C — General form: $AB + CD \rightarrow AD + CB$.

16. Combustion reactions always have which reactant?

A. Water

B. Nitrogen

C. Oxygen gas (O_2)

D. Carbon dioxide

✓ C — Combustion = substance + oxygen $\rightarrow$ oxides + energy.

17. A chemical equation is the best way to represent:

A. Only phase changes like melting

B. A chemical change where new substances form

C. Only physical mixing

D. Just a drawing of atoms

✓ B — It summarizes exactly what reacts, what forms, and their amounts and states.

18. If you see a symbol " Δ " written above the reaction arrow, it means:

A. Electricity is used

B. A catalyst is present

C. Heat is applied

D. Light is needed

✓ C — Delta (Δ) is the standard symbol for heating.

19. Which of the following is an example of a CHEMICAL change represented by an equation?

A. Sugar dissolving in tea

B. Cutting wood into pieces

C. Iron turning into rust

D. Ice melting to water

✓ C — Rust is a new substance; the others are only physical changes.

20. What is the main difference between a WORD EQUATION and a CHEMICAL EQUATION?

A. There is no difference

B. Chemical equations are always longer

C. Word equations show mass conservation; chemical equations do not

D. Word equations use only element names; chemical equations use formulas and symbols

✓ D — Example: "Hydrogen + Oxygen $\rightarrow$ Water" (word) vs. " $2 H_2 + O_2 \rightarrow 2 H_2O$ " (chemical).

21. When an insoluble solid forms and settles out of a solution during reaction, it is called a:

A. Product gas

B. Solvent

- C. Solute
- D. Precipitate

✓ D — Often marked (s) or "↓" in equations.

22. Which statement about subscripts is TRUE?

- A. They only matter for balancing
- B. You can change them to make an equation "look right"
- C. They are part of the chemical identity of a substance
- D. They count how many reactions happen

✓ C — Changing a subscript changes the compound itself (e.g., H_2O vs. H_2O_2).

23. In a chemical equation, coefficients apply to:

- A. Only the last element
- B. The entire formula they stand before
- C. Only the first element in the formula
- D. Nothing—they are just for counting

✓ B — "2 NaCl" means two whole units of sodium chloride, not two sodium atoms.

24. What information is NOT directly given in a typical chemical equation?

- A. Exact speed of the reaction
- B. Which substances are reactants and products
- C. Physical states of each substance
- D. Direction of the reaction

✓ A — Equations show what happens, not how fast it happens.

25. A reversible reaction (one that can go forward and backward) uses which symbol?

- A. ↓
- B. →
- C. $\rightleftharpoons$
- D. =

✓ C — Reactants form products, and products can reform reactants.

26. When balancing is NOT the focus, what does a chemical equation show clearly?

- A. The exact cost of chemicals
- B. Only the total mass
- C. The identity of reactants, products, and reaction type
- D. How dangerous it is

✓ C — Even unbalanced, it tells you what goes in, what comes out, and what kind of change occurs.

27. Which is TRUE about the arrow in a chemical equation?

- A. It separates reactants from products and shows direction of change
- B. It means "is equal to"
- C. It only appears in big reactions
- D. It means "can never reverse"

✓ A — It marks the boundary between starting materials and new substances.

28. If you see (g) next to a product, it means:

- A. It is hot
- B. It dissolves in water
- C. A gas is produced
- D. A solid is formed

✓ C — Common example: $\text{CO}_2(\text{g})$ or $\text{H}_2(\text{g})$.

29. Why do we use chemical formulas instead of full names in equations?

- A. Only for scientists
- B. To make it harder to read
- C. To show exactly which atoms and how many are present
- D. To make it shorter

✓ C — Formulas carry precise information about composition.

30. Which best describes the purpose of a chemical equation?

- A. To explain safety rules
- B. To summarize a chemical reaction in a standard, clear way
- C. To show how to balance numbers
- D. To list all lab equipment needed

✓ B — It is the universal language for describing chemical changes.

Group 6 · RATES OF CHEMICAL REACTIONS (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

Reaction rate measures how _____ a reaction happens.

- A. colorful
- B. hot

C. fast

D. big

✓ C — Change in concentration per unit time.

1. Higher temperature makes reactions generally:

A. Stop

B. Unaffected

C. Faster

D. Slower

✓ C — Faster particles = more/harder collisions.

2. Substance that speeds reaction without being consumed:

A. Inhibitor

B. Catalyst

C. Product

D. Reactant

✓ B — Lowers activation energy; enzymes are biological examples.

3. Reaction that RELEASES heat (feels hot):

A. Slow

B. Endothermic

C. Needs heating

D. Exothermic

✓ D — Exo = out; thermic = heat.

4. Reaction that ABSORBS heat (feels cold):

A. Endothermic

B. Fast always

C. Catalyzed

D. Exothermic

✓ A — Endo = in; takes heat from surroundings.

5. Smaller particle size → faster reaction because of larger:

A. Catalyst

B. Temperature

C. Surface area

D. Concentration

✓ C — More exposed area = more collisions.

6. Fire triangle = fuel + oxygen +:

A. Heat / ignition temperature

B. Water

C. Inhibitor

D. Catalyst

✓ A — Remove any one → fire stops.

7. Food stays fresh longer in fridge. Which factor?

A. Catalyst

B. Concentration

C. Surface area

D. Lower temperature slows decay

✓ D — #1 real-life food preservation application.

8. Substance that SLOWS reactions (food preservatives):

A. Product

B. Reactant

C. Inhibitor

D. Catalyst

✓ C — Opposite of catalyst; e.g., antioxidants.

9. Which is exothermic?

A. Burning wood / combustion

B. Melting ice

C. Cooking egg

D. Photosynthesis

✓ A — Combustion releases large amounts of heat.

10. Activation energy is:

A. Speed of reaction

B. Energy released

C. Total energy of reactants

D. Minimum energy needed to start reaction

✓ D — "Energy hill" reactants must climb.

11. Collision theory states reactions happen when particles:

A. Have high mass

- B. Are close only
- C. Touch lightly
- D. Collide with enough energy and correct orientation

✔ D — Effective collisions cause reaction.

12. Crushing a solid reactant:

- A. Adds catalyst
- B. Lowers concentration
- C. Increases surface area
- D. Lowers temperature

✔ C — Speeds reaction by exposing more particles.

13. Which factor does NOT increase rate?

- A. Higher concentration
- B. Catalyst
- C. Larger pieces
- D. Higher temperature

✔ C — Larger pieces = less surface area → slower.

14. Lowering temperature:

- A. Stops all reactions
- B. Speeds all reactions
- C. Has no effect
- D. Slows most reactions

✔ D — Particles move slower; fewer effective collisions.

15. Powdered CaCO_3 reacts FASTER than solid chips with acid because:

- A. Catalyst added
- B. Higher concentration
- C. Higher temperature
- D. Greater surface area

✔ D — Powder exposes far more particles to acid.

16. Concentrated acid reacts faster than dilute acid. Factor?

- A. Surface area
- B. Catalyst
- C. Higher concentration = more particles = more collisions
- D. Temperature

✔ C — More reactant in the same volume.

17. Enzymes in your body are examples of:

- A. Biological catalysts
- B. Products
- C. Inhibitors
- D. Reactants

✔ A — Speed digestion, respiration, etc.

18. Water extinguishes fire mainly by:

- A. Adding catalyst
- B. Removing fuel
- C. Increasing O_2
- D. Removing HEAT (cools below ignition temperature)

✔ D — Also smothers via steam blanket.

19. Fire blanket extinguishes fire by:

- A. Cooling
- B. Removing OXYGEN (smothers)
- C. Removing fuel
- D. Adding inhibitor

✔ B — Cuts off air supply.

20. Painting / galvanizing iron prevents rust by:

- A. Increasing concentration
- B. Adding catalyst
- C. Blocking contact with H_2O and O_2
- D. Increasing temperature

✔ C — Removes reactants needed for corrosion.

21. Which food preservation uses LOW water concentration (osmosis)?

- A. Freezing
- B. Refrigeration
- C. Canning
- D. Drying / salting / sugaring

✔ D — Removes water microbes need for decay reactions.

22. Canning food stops spoilage mainly by:

- A. Removing OXYGEN + killing microbes with heat
- B. Lowering temperature
- C. Increasing surface area
- D. Adding catalyst

✓ A — No O₂ = no aerobic decay reactions.

23. Crushing solid reactant into dust increases rate because:

- A. Adds catalyst
- B. Increases purity
- C. Increases surface area drastically
- D. Raises temperature

✓ C — Same mass, far more exposed particles.

24. Collision theory says reactions happen when particles:

- A. Stop moving
- B. Only touch lightly
- C. Are far apart
- D. Collide with correct orientation AND enough energy

✓ D — Effective collisions = reaction; others bounce.

25. Stirring reactants usually:

- A. Stops reaction
- B. Changes product
- C. Speeds reaction
- D. Slows reaction

✓ C — Brings fresh reactants together faster.

26. Which is endothermic?

- A. Neutralization
- B. Burning
- C. Freezing water
- D. Photosynthesis

✓ D — Needs continuous energy input from sunlight.

27. Why does higher temperature INCREASE rate so much?

- A. Lowers surface area
- B. Changes reactants
- C. More frequent collisions AND more have energy $\geq$ activation energy
- D. Only more collisions

✓ C — Energy factor is bigger than frequency.

28. Catalyst speeds reaction by:

- A. Becoming product
- B. Lowering required activation energy
- C. Increasing temperature
- D. Increasing surface area

✓ B — Makes the "energy hill" easier to climb.

29. Photosynthesis needs constant sunlight. It is:

- A. Very fast
- B. Catalyst free
- C. Endothermic
- D. Exothermic

✓ C — Continuously absorbs light energy.

30. Explosions are extremely fast due to:

- A. Instant gas release + heat + huge surface contact
- B. Inhibitors
- C. Low concentration
- D. Low temperature

✓ A — All rate factors maximized at once.

31. Reaction fast at start, slows, stops. Why?

- A. Catalyst disappears
- B. Temperature increases
- C. Surface area grows
- D. Reactants used up → concentration drops to zero

✓ D — No reactant particles left = no more collisions.

32. How does a catalyst reduce air pollution in car exhaust?

- A. Blocks exhaust
- B. Speeds conversion of toxic gases → harmless N₂ + CO₂ + H₂O
- C. Burns fuel
- D. Cools engine

✓ B — Catalytic converters = environmental catalyst application.

33. Higher O₂ concentration makes fire burn faster. This is which factor?

- A. Temperature
- B. Catalyst
- C. Concentration of reactant (O₂)
- D. Surface area

✓ C — More O₂ molecules = more frequent effective collisions.

34. The MAIN factor in corrosion prevention (paint/zinc) is:

- A. Eliminate contact between metal and O₂/H₂O
- B. Increase concentration
- C. Add catalyst
- D. Lower temperature

✓ A — Remove one required reactant → reaction cannot happen.

Group 7 · HOMEOSTASIS (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

1. Homeostasis is best defined as:

- A. Constant unchanging state
- B. Only temperature control
- C. Stable internal balance despite external changes
- D. All body reactions stopping

✓ C — Dynamic balance; conditions fluctuate within a normal range.

2. Which organ system is the MAIN control center for homeostasis?

- A. Skeletal system
- B. Digestive system
- C. Excretory system
- D. Nervous + Endocrine systems

✓ D — Nerves send fast signals; hormones give slower, longer lasting control.

3. When body temperature rises, you sweat to cool down. This is:

- A. No feedback
- B. Negative feedback
- C. Positive feedback
- D. Stimulus only

✓ B — Reverses the change back to normal.

4. The “normal” or target value in homeostasis is called the:

- A. Set point
- B. Effector
- C. Response
- D. Stimulus

✓ A — E.g., human body temperature set point ≈ 37°C.

5. Which detects changes in the body’s internal or external environment?

- A. Control center
- B. Response
- C. Receptor / Sensor
- D. Effector

✓ C — E.g., nerve endings in skin detect heat/cold.

6. Which part carries out the action to fix the change?

- A. Stimulus
- B. Effector (muscle / gland)
- C. Receptor
- D. Control center

✓ B — Muscles contract; glands release hormones/sweat.

7. Negative feedback works to:

- A. Reverse / correct change back to set point
- B. Stop all activity
- C. Change the set point permanently
- D. Increase change

✓ A — Most common homeostatic mechanism.

8. Which is NOT maintained by homeostasis?

- A. Blood glucose
- B. Random constant height
- C. Blood pH
- D. Body temperature

✓ B — Height changes with growth; not a regulated balance.

9. The control center for temperature regulation is the:

- A. Heart

B. Lungs

C. Hypothalamus (brain)

D. Liver

✓ C — Receives signals and sends commands.

10. When you are cold, blood vessels in skin:

A. Burst

B. Constrict / narrow

C. Dilate / widen

D. Disappear

✓ B — Keeps warm blood inside core → reduces heat loss.

11. Which system uses chemical messengers called hormones?

A. Digestive

B. Nervous

C. Endocrine

D. Circulatory

✓ C — Slower but longer lasting control.

12. Insulin and glucagon regulate:

A. Water balance

B. Blood glucose level

C. Heart rate

D. Body temperature

✓ B — Work together to keep sugar stable.

13. If blood sugar rises, pancreas releases:

A. Thyroxine

B. Adrenaline

C. Insulin

D. Glucagon

✓ C — Lowers sugar by moving it into cells.

14. If blood sugar drops, pancreas releases:

A. Estrogen

B. Insulin

C. Cortisol

D. Glucagon

✓ D — Releases stored sugar from liver.

15. Kidneys help maintain:

A. Water and salt balance

B. Bone density

C. Muscle strength

D. Body temperature

✓ A — Adjust urine concentration.

16. Shivering when cold is an example of:

A. Exothermic reaction only

B. No homeostatic control

C. Negative feedback generating heat

D. Positive feedback increasing heat loss

✓ C — Muscle movement produces heat to raise body temperature back to the set point.

17. Which is NOT a key homeostatic balance in humans?

A. Water / salt balance

B. Constant body weight forever

C. Blood glucose level

D. Body temperature

✓ B — Weight changes naturally; it is not a fixed regulated set point.

18. Insulin lowers blood sugar; glucagon raises it. This is:

A. Positive feedback loop

B. Excretion only

C. Opposing negative feedback

D. Uncontrolled reaction

✓ C — Two hormones work against each other to keep glucose stable.

19. Positive feedback differs from negative feedback because it:

A. Has no control center

B. Only works in plants

C. Reverses the change

D. AMPLIFIES / increases the change away from the set point

✓ D — Rare in the body; e.g., childbirth, blood clotting.

20. When blood sugar rises after eating, the pancreas releases:

- A. Adrenaline
- B. Glucagon
- C. Thyroxine
- D. Insulin

✓ D — Insulin helps cells take in glucose; lowers blood sugar.

21. Which organ MAINLY controls water balance and waste removal?

- A. Stomach
- B. Heart
- C. Kidney
- D. Lungs

✓ C — Adjusts urine concentration; keeps salt/water levels stable.

22. Sweating and vasodilation both help:

- A. Raise blood sugar
- B. Lower body temperature
- C. Lower blood pressure
- D. Raise body temperature

✓ B — Remove excess heat from the body.

23. Which is an example of positive feedback?

- A. Insulin action
- B. Sweating
- C. Blood clotting
- D. Breathing rate

✓ C — More clotting chemicals → faster clotting until sealed.

24. Homeostasis requires:

- A. Constant energy input
- B. No energy
- C. Only sleep
- D. Only food

✓ A — Active balancing process.

25. If homeostasis fails for blood pH, this happens first:

- A. Faster breathing
- B. More sweating
- C. Growth
- D. Enzymes stop working / denature

✓ D — Wrong pH changes protein shape.

26. When you exercise, breathing rate increases to balance:

- A. Heat in / heat out
- B. Water in / salt out
- C. O₂ in / CO₂ out
- D. Sugar in / fat out

✓ C — Maintain gas balance in blood.

27. The order of a feedback loop is:

- A. Control → Stimulus → Receptor → Effector
- B. Response → Stimulus → Effector → Control
- C. Stimulus → Receptor → Control → Effector → Response
- D. Effector → Receptor → Control → Response

✓ C — Correct sequence for all homeostatic loops.

28. Which is the ONLY example of POSITIVE feedback below?

- A. Kidneys saving water when dehydrated
- B. Sweating when hot
- C. Insulin lowering sugar
- D. Oxytocin increasing stronger contractions during birth

✓ D — More contraction → more hormone → even stronger contraction until birth.

29. If homeostasis FAILS for a long time, the result is:

- A. Faster growth
- B. Permanent set point change
- C. Disease / organ damage / death
- D. Improved health

✓ C — Unbalanced conditions damage cells and stop organs working.

30. Which statement about homeostasis is TRUE?

- A. Only controls temperature
- B. Stops when you sleep
- C. Only happens in warm-blooded animals
- D. Requires constant energy input

✓ D — Active process; needs energy to keep balancing levels.

31. Dehydration causes kidneys to:
- A. Reabsorb more water back into blood
 - B. Release more water
 - C. Make more urine
 - D. Stop working
- ✓ A — Negative feedback restores water balance.
32. When blood pressure rises, body responds to:
- A. Stop heart
 - B. Ignore it
 - C. Lower it back to normal
 - D. Raise it more
- ✓ C — Negative feedback corrects the change.
33. Which best describes “dynamic equilibrium”?
- A. Rapid wild swings
 - B. Never changes
 - C. Always exactly the same
 - D. Small changes within a healthy range
- ✓ D — Homeostasis is stable but flexible.
34. Diabetes is a disease related to which homeostatic failure?
- A. Water balance
 - B. Blood glucose regulation
 - C. Salt balance
 - D. Temperature
- ✓ B — Problem with insulin production or use.
35. Without homeostasis:
- A. No change occurs
 - B. Growth speeds up
 - C. Cells and organs cannot function
 - D. Body works better
- ✓ C — Conditions drift outside safe limits → damage or death.

Group 8 · MECHANISMS OF EVOLUTION (only use 15 questions in every pretest topic, this will be rephrased and randomized each playthrough)

1. Evolution is BEST defined as:
- A. Instant creation of new species
 - B. Only large animals getting stronger
 - C. Gradual change in the inherited traits of a POPULATION over many generations
 - D. Individual organisms changing during their lifetime
- ✓ C — Evolution happens to populations, not single individuals; traits are passed across generations.
2. Who is the scientist who first proposed natural selection as the main mechanism of evolution?
- A. Charles Darwin
 - B. Gregor Mendel
 - C. Antoine Lavoisier
 - D. Albert Einstein
- ✓ A — Darwin published On the Origin of Species in 1859 based on Galápagos Islands observations.
3. Natural selection is often summarized as:
- A. Traits you acquire during life are passed to offspring
 - B. Survival and reproduction of organisms BEST FITTED to their environment
 - C. Only the largest animals survive
 - D. Strongest always kill the weakest
- ✓ B — “Fitness” = ability to survive AND produce fertile offspring, not just physical strength.
4. Any inherited trait that helps an organism survive and reproduce in its environment is called:
- A. Adaptation
 - B. Variation
 - C. Fossil
 - D. Isolation
- ✓ A — Example: thick fur in polar bears, long neck in giraffes.
5. The natural differences in traits between individuals of the same species (e.g., different beak sizes in finches) is called:
- A. Selection
 - B. Heredity
 - C. Variation
 - D. Isolation
- ✓ C — Variation must already exist in a population for natural selection to act on it.

6. For evolution to happen, useful traits must be able to be passed from parents to offspring. This is called:

- A. Isolation
- B. Variation
- C. Adaptation
- D. Heredity / Inheritance

✓ D — Traits only affect evolution if they are genetic and can be inherited.

7. Preserved remains, traces, or impressions of ancient organisms found in rock layers are called:

- A. Endemic species
- B. Fossils
- C. Adaptations
- D. Homologous structures

✓ B — Examples: bones, shells, footprints, leaf imprints trapped in sedimentary rock.

8. Body parts of different species that have SIMILAR BONE STRUCTURE but different functions (e.g., human arm, whale flipper, bat wing) are:

- A. Vestigial structures
- B. Fossils
- C. Homologous structures
- D. Analogous structures

✓ C — Same basic plan = inherited from a common ancestor.

9. A body part that has lost most or all of its original function through evolution (e.g., human appendix, tailbone) is:

- A. Analogous
- B. Adaptation
- C. Homologous
- D. Vestigial structure

✓ D — It is reduced in size; remains as evidence of an ancestor that used it fully.

10. The study of where different species live on Earth and why they are found there is:

- A. Genetics
- B. Comparative anatomy
- C. Paleontology
- D. Biogeography

✓ D — Explains why islands often have unique species found nowhere else.

11. The peppered moth in England changed from mostly light to mostly dark during the Industrial Revolution because tree trunks turned black with soot. This is a famous example of:

- A. Biogeography
- B. Artificial selection
- C. Natural selection in action
- D. Vestigial formation

✓ C — Dark moths survived predation by birds better; they reproduced more and became common.

12. A river changes course and permanently splits one population of beetles into two groups that never meet again. This is:

- A. Natural selection
- B. Artificial selection
- C. Reproductive isolation
- D. Geographic isolation

✓ D — Physical barrier (river, mountain, ocean) separates populations; first step to forming new species.

13. Bird wing and insect wing have the SAME FUNCTION (flying) but completely different internal structure. They are:

- A. Vestigial structures
- B. Homologous structures
- C. Fossils
- D. Analogous structures

✓ D — Same job, different origin = evolved independently, NOT from a common flying ancestor.

14. Why is isolation important in forming NEW species (speciation)?

- A. It creates fossils faster
- B. It increases mutation rate instantly
- C. It stops gene flow between groups, letting them evolve differently over time
- D. It makes organisms weaker

✓ C — Without contact, separated populations accumulate different traits until they can no longer interbreed.

15. Fossils found in DEEPER / older rock layers are generally:

- A. Simpler and more different from living species today
- B. Only large mammals
- C. Identical to modern organisms
- D. More complex than modern life

✓ A — Fossil record clearly shows simple life first, then gradual change to more complex forms over time.

16. Which is the BEST evolutionary explanation for giraffes' long necks?

- A. Long-necked variants survived better, reproduced more, and passed the trait on
- B. A random mutation made all necks long in one generation

- C. They stretched to reach high leaves and passed stretched necks to babies
- D. All giraffes chose to grow longer necks

✓ A — Variation existed first; natural selection favored longer necks over many generations.

17. Farmers choosing only the largest, sweetest mangoes to plant next season is:

- A. Geographic isolation
- B. Artificial selection
- C. Vestigial selection
- D. Natural selection

✓ B — Humans, not nature, select which traits get passed on.

18. Early embryos of fish, chicken, pig, and human look very similar. This evidence of common ancestry comes from:

- A. DNA sequencing only
- B. Comparative embryology
- C. Fossil record
- D. Biogeography

✓ B — Shared early developmental stages = inherited from a shared vertebrate ancestor.

19. Two groups of frogs live in the same pond but mate at different months of the year and never interbreed. This is:

- A. Geographic isolation
- B. Analogous breeding
- C. Reproductive isolation
- D. Artificial selection

✓ C — No physical barrier, but differences in behavior/time prevent mating; they are now separate species.

20. The Philippine Eagle is found only in the Philippines and nowhere else on Earth. It is called:

- A. An endemic species
- B. An artificial species
- C. A vestigial species
- D. A fossil species

✓ A — Biogeography: islands and isolated countries evolve many unique endemic species.

21. Which is the CORRECT sequence of events for a new species to form (speciation)?

- A. Isolation → New species → Variation → Selection
- B. New species → Variation → Isolation → Selection
- C. Variation → Geographic Isolation → Natural Selection → Reproductive Isolation → New species
- D. Selection → Variation → Isolation → New species

✓ C — Variation must exist first; separation stops gene flow; selection makes groups different until they can no longer interbreed.

22. Why is natural selection considered the PRIMARY mechanism of evolution, not mutation, genetic drift, or chance alone?

- A. It was the first mechanism discovered
- B. It is the only mechanism that CONSISTENTLY produces ADAPTATIONS matched to the environment
- C. It is the only mechanism that changes traits
- D. It works instantly in one generation

✓ B — Other mechanisms exist, but only natural selection reliably builds useful, environment-matched traits over time.

23. Modern whales have tiny, unused pelvic (hip) bones buried deep inside their bodies. This vestigial structure is strong evidence that:

- A. Whales evolved from fish directly
- B. The pelvic bone has always been useless
- C. Whales are still evolving legs today
- D. Whales evolved from 4-legged land-dwelling mammals that moved back to the sea

✓ D — Fossil series confirms this: walking whale → semi-aquatic → fully aquatic modern whale.

24. Someone says: “Evolution is just a theory, so it’s just a guess.” The CORRECT scientific response is:

- A. In science, “theory” means a well-tested explanation supported by massive, repeated evidence from many fields
- B. Evolution is only a hypothesis
- C. Theories become laws once proven; evolution is almost a law
- D. True, it has no evidence yet

✓ A — Scientific theory ≠ casual “guess”; it is the highest, most reliable level of scientific explanation.

25. Fossils of the same extinct reptile Mesosaurus are found only in eastern South America and western Africa. This, plus matching coastlines, strongly supports:

- A. It evolved independently on both continents
- B. Continental drift / Pangaea: these continents were once joined together
- C. The species swam across the Atlantic Ocean
- D. Fossils were carried by humans

✓ B — Key biogeography evidence: landmasses split and carried their resident species with them.

26. Which of the following is NOT a required condition for natural selection to occur?

- A. Traits acquired during an organism's life (e.g., bigger muscles from exercise) are passed to offspring
- B. Variation exists in the population
- C. Traits are heritable
- D. More offspring are born than can survive

✓ A — This was Lamarck's WRONG idea; only genetic/inherited traits, not acquired ones, are passed on.

27. Homologous structures (e.g., forelimbs of vertebrates) are strong evidence of common ancestry mainly because:

A. They perform exactly the same function

B. They appear only in fossil species

C. They share the same unique bone arrangement inherited from a shared ancestor, modified for different jobs

D. They look identical externally

✓ C — Same basic plan, different uses = descent with modification from one ancestor.

28. A severe 10-year drought hits an island; all small, soft seeds die, leaving only large, hard seeds. Over generations, what will natural selection do to the island's finch population?

A. All finches will instantly grow larger beaks in one year

B. Average beak size will become LARGER, thicker, and stronger

C. Average beak size will get smaller and weaker

D. Beak size will not change at all

✓ B — Birds with larger, stronger beaks survive and reproduce; the trait becomes more common over generations.

29. If two separated populations are NOT geographically isolated and individuals keep moving/mating between them:

A. Mutations stop occurring

B. Natural selection cannot happen at all

C. Gene flow continues; they will NOT diverge into separate species

D. They will definitely become separate species

✓ C — Constant mixing keeps their gene pools similar; isolation is required for populations to split into new species.

30. Which combination of evidence gives the STRONGEST, most reliable support for evolution by common descent?

A. Only DNA similarity alone

B. Only embryology alone

C. Fossil record + homologous structures + DNA/protein similarity + biogeography, all agreeing

D. Only fossil record alone

✓ C — Multiple independent lines of evidence all converging on the same conclusion is what makes evolution one of the most strongly supported theories in all of science.